

α -tempered Variational Bayes Calibration over Sparse Variational Gaussian Process

Statistics Capstone Project

Yifei Yang

Mentors: Dr. Meng Li^{}, Dr. Elizabeth McGuffey[†], Jiaming Liu[‡] and Veronica Patterson[§]*

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Abstract. Spatial prediction models are useful in many applications, such as interpolating pollution levels at unsampled locations given a sample of PM_{2.5} readings. Kriging, or noiseless Gaussian process, is a type of spatial prediction model. It provides both point predictions and uncertainty bands, but these bands often suffer from under coverage, especially in case of model mismatch or when scaling methods are used. This project proposes an α -tempered variational Bayes (TVB) version of sparse variational Gaussian process (SVGP) that introduces a single temperature parameter, α , on the likelihood to calibrate uncertainty while retaining scalability. We first evaluate coverage using synthetic data from a known true function f_0 . We observe under-coverage of the baseline untempered case ($\alpha = 1$) and also a monotonic coverage increase when tempering down α values ($\alpha < 1$). Next, we calibrate 100 different α values using bootstrapping and we find the α values that reach near-nominal coverage over the full test grid with minimal loss in mean prediction accuracy. The results of this project are a clear and accessible pipeline for scalable Kriging with calibrated near-nominal uncertainty.

^{*}Noah Harding Associate Professor, Rice University, Department of Statistics

[†]Associate Teaching Professor, Rice University, Department of Statistics

[‡]Ph.D. Candidate, Rice University, Department of Statistics

[§]Ph.D. Student, Rice University, Department of Statistics

1. Introduction

Spatial prediction models enable us to predict unobserved locations based on nearby measurement data. They are used extensively in areas such as environmental mapping, computer experiments, and robotics to enable better risk-based decision making [1, 2]. One application example is urban air-quality mapping, in which spatial prediction models interpolate pollution readings from a sparse sensor network across a city, yielding a continuous concentration map with location-specific uncertainty for exposure assessment and for siting new monitors. Another practical example is surrogate modeling for computer experiments: In costly simulators (e.g., aerodynamics or hydrology), spatial prediction models learn from a small set of runs to predict outcomes at new settings. They provide fast, map-like predictions over space or design choices, plus uncertainty estimates to guide where to test next.

At the statistical level, spatial prediction is typically formalized using Kriging methods, also known as noiseless Gaussian process (GP) models. These methods treat the underlying spatial process as a continuous but unknown surface and consider the values at different locations as correlated random variables. Assuming that nearby points are more correlated than distant ones, the model can "borrow" information from neighboring observations when predicting unsampled locations. This notion of spatial similarity is encoded by a kernel (covariance) function $k(x, x')$, which maps the distance (and possibly direction) between locations to an expected level of correlation, thereby controlling how rapidly information decays across space. This probabilistic framework not only generates a smooth estimated surface but also quantifies the uncertainty at each location, providing higher confidence in densely monitored areas and wider confidence intervals in sensor-sparse regions [1, 2]. In the example of air quality maps, these uncertainty estimates are crucial for identifying high-risk areas and prioritizing the placement of new sensors.

However, there are several limitations in Kriging applications. Firstly, classical Kriging is not compatible with large data. GP inference factors an $n \times n$ covariance matrix, where n is the number of data points. This results in $\mathcal{O}(n^3)$ computation time and $\mathcal{O}(n^2)$ memory, which is both time-consuming and expensive, especially when dealing with large datasets [2]. Prior works such as the sparse variational Gaussian process (SVGP) provide practical scalability improvements by introducing $m \ll n$ inducing points, which remarkably reduces the computation time to only $\mathcal{O}(nm^2 + m^3)$ [3]. Methodologically, SVGP frames inference as a variational approximation to the full GP: positing inducing variables u at locations Z and optimizing a tractable variational posterior $q(u)$ via an evidence lower bound (ELBO), which in turn induces approximate posteriors and predictions at arbitrary locations [3, 4]. Additionally, SVGP supports mini-batch training and GPU acceleration for large-scale ap-

plications [5].

Another limitation is that uncertainty measurements for Kriging predictions are often unreliable. In common practical situations, prediction intervals could easily become too narrow (overconfident) or too wide (under-informative)—for example, using a stationary kernel for a non-stationary field, incorrect specification of noise, or sparse/uniform sampling [2]. The standard variational posterior distribution has the issue of insufficient dispersion, meaning it can underestimate posterior uncertainty and yield prediction intervals that are systematically too narrow, reducing empirical coverage [4]. In practice, this creates a false sense of precision: forecasts appear confidently concentrated even when the model is misspecified or data are sparse, which makes uncertainty-sensitive decisions unreliable, complicates reproducibility across datasets or replications, and weakens the interpretability of uncertainty bands as meaningful measures of risk. To address these issues, recent researchers have provided partial solutions: SVGP for scalability, with emerging contraction-rate guarantees [6], and generalized or α -tempered updates that reweight the likelihood relative to the prior to calibrate posterior spread—widening too-narrow intervals or shrinking too-wide ones [7–9]. Still, a gap remains for further research: there is currently no simple and scalable pipeline that applies tempering in SVGP to achieve a latent-function coverage close to the nominal level while maintaining its scalability through the inducing-points mechanism.

In this project, we use an α -tempered sparse variational model to fix how “wide” our uncertainty band is for the latent surface f . We start with the standard SVGP setting ($\alpha = 1$) on simulated data where the truth is known and confirm undercoverage: the prediction bands are too narrow (they miss the truth too often) on a fixed set of test locations. We then turn a single knob, $\alpha \in (0, 1]$, which down-weights the data likelihood relative to the prior; smaller α widens the bands without sacrificing reasonable mean accuracy [3, 4]. We choose α^* by a simple calibration rule: pick the smallest value whose empirical coverage is close to the nominal target on the entire test grid. Implementation details follow the SVGP tools of GPyTorch [5] to achieve efficient GPU training, and SVGP is both scalable and supported by emerging theoretical guarantees [6].

The paper is organized as follows. Section 2 introduces the synthetic data, the standardization routine, and the training configurations. Section 3 presents the statistical methods, modeling framework, and the replication pipeline used for coverage assessment. Section 4 reports the empirical coverage assessment results and their interpretation. Section 5 then introduces the calibration method and pipeline. Section 6 follows with an interpretation of the calibration results. Finally, Section 7 offers a discussion and outlines directions for future work.

A complete list of notation used throughout the paper is provided in Appendix [subsection A.2](#).

2. Data Description.

Overview All data are synthetic. Data are generated over a chosen test range from a known true function f_0 . Because f_0 is known, we can measure frequentist coverage without ambiguity.

True Function (Figure 1). We choose

$$f_0(x) = \sin(2\pi x) + 0.5 \cos(3x).$$

choose this function because it is analytic, bounded, and exhibits multiple characteristic lengthscales, which makes it informative for assessing coverage under single-lengthscale radial basis function (RBF) kernels, i.e. the standard squared-exponential Gaussian kernel $k(x, x') = \sigma_f^2 \exp\left(-\frac{(x-x')^2}{2\ell^2}\right)$.

The input domain is the symmetric interval $[-2.5, 2.5]$, so the empirical mean of x is near 0 and the standard deviation is close to that of $\text{Uniform}(-2.5, 2.5)$, i.e., $\sqrt{5^2/12} \approx 1.443$.

Raw Test Inputs (fixed grid). Coverage is evaluated on a *fixed, uniform grid* over $[-2.5, 2.5]$. We take $n_{\text{test}} = 50$ equally spaced points to form X_{test}^* , and each test point is drawn by:

$$x_j^* = -2.5 + \frac{j-1}{J-1} \cdot 5, \quad j = 1, \dots, n_{\text{test}}.$$

Training targets are noise-free:

$$y_i = f_0(x_i).$$

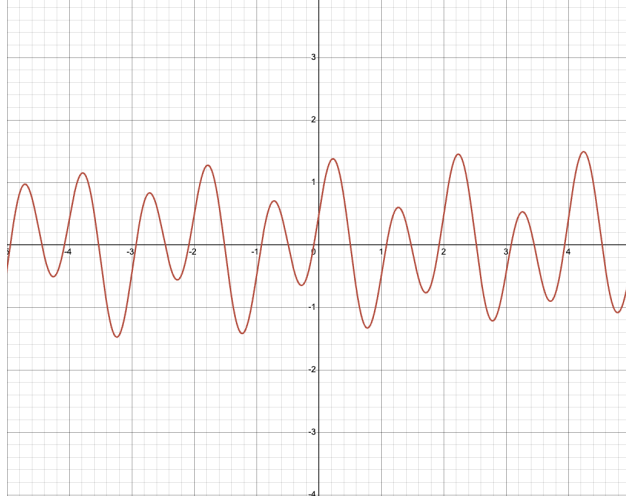


Figure 1: True function $f_0(x) = \sin(2\pi x) + 0.5 \cos(3x)$. Multiple frequency components are presented, making it suitable for coverage analysis.

Raw Training Inputs. We use X_{train}^* with size of $n_{\text{train}} = 500$ on $[-2.5, 2.5]$ generated via stratified *Latin-hypercube sampling* (LHS) design. LHS partitions the interval into 500 equal strata, draws one uniform point from each stratum, and then randomly permutes all training points.

Raw Inducing Points (SVGP). We use $M = 50$ inducing points. The inducing points Z^* are sampled as 50 equally spaced locations over $[-2.5, 2.5]$:

$$z_m^* = -2.5 + \frac{m-1}{M-1} \cdot 5, \quad m = 1, \dots, M.$$

Training Inputs Standardization. To eliminate the data discrepancies between different settings, we apply *symmetric linear normalization* process to all the inputs—training points X_{train} , test points X_{test} , and inducing locations Z —that maps the raw domain $[L, H] = [-2.5, 2.5]$ to $[-1, 1]$. Define

$$m = \frac{L+H}{2}, \quad h = \frac{H-L}{2}, \quad x_{\text{std}} = \frac{x_{\text{raw}} - m}{h}.$$

The same transform is then applied componentwise to each set:

$$X_{\text{train}} = \frac{X_{\text{train}}^* - m}{h}, \quad X_{\text{test}} = \frac{X_{\text{test}}^* - m}{h}, \quad Z = \frac{Z^* - m}{h}.$$

This single transformation is fixed in the initial state (from $[L, H]$) and is repeatedly used in all replications and the calculation of all α values, thereby maintaining the geometric

structure unchanged, improving the numerical conditions, and eliminating the preprocessing differences between each repetitive operation. Because the raw training set X_{train}^* is generated *once* via LHS design and then held fixed, the locations do not change across epochs; during optimization we therefore only shuffle the order of X_{train} each epoch to randomize minibatch composition.

Training Schedule. Each replication runs 1500 epochs. With mini-batches of 500, each epoch processes 1 batch that include all training data.

Predictive Setup (Details for Reproduction). We use a constant mean with an RBF kernel wrapped in a `ScaleKernel` and a Gaussian likelihood, and we train with the Adam optimizer (learning rate 2×10^{-2}), FP64 precision, jitter 10^{-2} , and per-epoch logging [5]. By using the normal quantiles, we will check the coverage range of the SVGP posterior distribution for $f(X_{\text{test}}^*)$, and the overall coverage is the average of all these point estimates.

This setup fixes the raw training design X_{train}^* (used in standardized form X_{train}) and the test grid X_{test}^* , enabling a fair, pointwise coverage assessment of the latent posterior $f(X_{\text{test}}^*)$.

Data Summary. Table 1 summarizes all the simulation and training configurations.

Table 1: Simulation and training configuration.

Quantity	Value
Training size n_{train}	500
Test size n_{test}	50 (uniform grid on $[-2.5, 2.5]$)
Replications R	100
Truth $f_0(x)$	$\sin(2\pi x) + 0.5 \cos(3x)$
Training design	Stratified (LHS) on $[-2.5, 2.5]$
Evaluation target	Latent $f(X_{\text{test}}^*)$ (noise-free)
SVGP inducing points m	50 (learned locations)
Epochs / batch size	1500 / 500
Optimizer / LR	Adam / 2×10^{-2}
Kernel	RBF with scale kernel (learned)
Likelihood	GaussianLikelihood (fixed noise / nugget = 3×10^{-3} ; latent- f evaluation)
Precision / jitter	FP64, jitter 10^{-2}
Software	PyTorch 2.5.1, GPyTorch 1.14 (SVGP implementation)

3. Coverage Assessment and α -Tempered SVGP

We used the Python package *GPyTorch*, which provides existing functions for SVGP model training, following the standard SVGP formulation [2, 3]. The existing code structures an SVGP model using contour integral quadrature (CIQ) with msMINRES [5], which accelerates matrix-vector multiplications and enables scalable, GPU-efficient inference for large kernel matrices. The output of the GPyTorch standard SVGP model includes mean and (co)variance for the latent function. We will extract the mean and standard deviation from the model to form 95% credible interval bands. Our addition to the existing code is an orchestration layer that repeats training over R independent Latin-hypercube sampled (LHS) designs, with per-replication standardization and inducing-point initialization. With our pre-constructed coverage matrix from latent-function credible intervals, we are able to estimate point-wise and overall frequentist coverage by outputting 0/1 inclusion indicators across runs.

3.1. Model

The model places a Gaussian process prior on the latent function f with zero mean and a RBF kernel, where the hyperparameters are $\theta = (\ell, \sigma_f^2)$.

$$f \sim GP(0, k_\theta(\cdot, \cdot)), \quad k_\theta(x, x') = \sigma_f^2 \exp\left(-\frac{(x-x')^2}{2\ell^2}\right)$$

To scale inference to large n_{train} , we adopt the SVGP framework [2, 3, 5]: introduce $m = 50$ inducing inputs $Z = \{z_j\}_{j=1}^m$ and inducing variables $u = f(Z)$, posit a variational posterior $q(u) = \mathcal{N}(\mu, S)$, and optimize $\{\mu, S, \theta, Z\}$ by maximizing a tractable lower bound to the log marginal likelihood. Here, f denotes the noise-free regression function (with f_0 used only to generate synthetic data for evaluation), ℓ is the kernel length scale, σ_f^2 is the signal variance, Z is the (learned) inducing locations, and μ and S are the mean vector and covariance matrix of $q(u)$.

3.2. Model assumptions and checks

We assume smoothness compatible with the RBF prior and homoskedastic (constant-variance) Gaussian observation noise. The repeated experiments are independent of each other, and each repetition uses a brand-new LHS design. When using FP64 floating-point operations and with a jitter value of 10^{-2} , the ELBO can stabilize at a plateau within 20 cycles. We monitor the ELBO trajectory, the learned length scale, the output scale, and the noise variance, and report the diagnostic results as health check indicators, rather than target indicators, at the end of the run.

3.3. Objective (ELBO and tempering)

The optimization objective in the SVGP framework is the standard evidence lower bound

$$\text{ELBO} = \mathbb{E}_{q(f)}[\log p(y | f)] - \text{KL}(q(u) \| p(u)).$$

Maximizing the ELBO is equivalent (within the chosen variational family) to minimizing the Kullback–Leibler (KL) divergence, the second term in the ELBO equation; When the ELBO reaches a (local) maximum, the variational posterior $q(u)$ provides a good *within-family* approximation to the exact posterior $p(u | y)$. This typically improves predictive fit, but it does not, by itself, guarantee calibrated uncertainty or recovery of the data-generating distribution.

We now introduce an α -tempered (generalized Bayes) posterior,

$$p_\alpha(f | y) \propto p(f) p(y | f)^\alpha, \quad 0 < \alpha \leq 1,$$

which down-weights the likelihood relative to the prior and is often used to adjust posterior dispersion for coverage calibration [4, 7–9]. Equivalently, the α -tempered ELBO multiplies the expected log-likelihood term by α :

$$\mathcal{L}_\alpha = \alpha \mathbb{E}_{q(f)}[\log p(y | f)] - \text{KL}(q(u) \| p(u)).$$

When $\alpha = 1$, this reduces to the standard ELBO baseline. In the SVGP code, α -tempering is implemented simply by setting $\beta = 1/\alpha$ in the MLL function.

3.4. Two-stage training procedure

Through replications, a two-stage procedure was utilized to maintain the accuracy of training. Tempering the likelihood improves calibration by inflating posterior spread but commonly trades some efficiency for validity [10]. We therefore fit the mean structure at $\alpha = 1$ and calibrate uncertainty by training only the posterior spread. In stage 1, we trained the model under standard SVGP settings where $\alpha = 1$. This stage learns the mean and kernel hyperparameters and provides a well-behaved baseline fit of the latent surface f . In stage 2, the Stage-1 model is reloaded, for $\alpha \in (0, 1]$, updating only the variational covariance while holding the mean and kernel learned from stage 1 fixed. This method isolates the effect of α -tempering on coverage while avoiding additional stochasticity caused by loss of accuracy when α is reduced.

3.5. Posterior prediction for the latent function

After optimizing the model by maximizing the ELBO, we obtain the variational posterior over the inducing variables $u = f(Z)$ as $q(u) = \mathcal{N}(m, S)$. The latent predictive distribution

at the fixed test grid $X^* = \{x_j^*\}_{j=1}^J$ is then given by marginalizing out inducing variables u :

$$q(f^*) = \int p(f^* | u) q(u) du,$$

where $p(f^* | u)$ is the GP conditional on u . Let $Z = \{z_i\}_{i=1}^m$ be the inducing inputs and the kernel blocks

$$K_{ZZ} = k(Z, Z), \quad K_{X^*Z} = k(X^*, Z), \quad K_{ZX^*} = k(Z, X^*), \quad K_{X^*X^*} = k(X^*, X^*).$$

Evaluating the integral yields the standard SVGP predictive posterior [3]:

$$\begin{aligned} q(f^*) &= \mathcal{N}(\mu^*, \Sigma^*), \\ \mu^* &= K_{X^*Z} K_{ZZ}^{-1} \mu, \\ \Sigma^* &= K_{X^*X^*} - K_{X^*Z} (K_{ZZ}^{-1} - K_{ZZ}^{-1} S K_{ZZ}^{-1}) K_{ZX^*}. \end{aligned}$$

For each grid location x_j^* , we form the 95% credible interval for the latent function as

$$\text{CI}_{0.95}(x_j^*) = \mu^*(j) \pm 1.96 \sqrt{\Sigma^*(j, j)}.$$

3.6. Coverage matrix and aggregation

For replication and coverage assessment, we construct a matrix $H \in \{0, 1\}^{R \times J}$. Here R is the number of replications ($R = 100$) and J is the number of test locations ($n_{\text{test}} = 50$), so the matrix is of size 100×50 .

Using the computed 95% credible bands, we define

$$H_{rj} = \mathbf{1} \left\{ f_0(x_j^*) \in \text{CI}_{0.95}^{(r)}(x_j^*) \right\},$$

which indicates whether, at the j -th test point in the r -th replication, the credible band covers the true value $f_0(x_j^*)$. Each row of H records coverage outcomes from one independently trained SVGP, while each column aggregates coverage at a fixed location across replications. The pointwise coverage curve is the column mean

$$\widehat{C}(j) = \frac{1}{R} \sum_{r=1}^R H_{rj},$$

visualized against x_j^* , and the overall latent- f coverage is the grand mean

$$\overline{C} = \frac{1}{RJ} \sum_{r=1}^R \sum_{j=1}^J H_{rj}.$$

Both $\widehat{C}(j)$ and $\overline{C}$ lie in the interval $[0, 1]$ and represent coverage levels in decimal form.

3.7. Software

Implementation uses PyTorch 2.5.1 and GPyTorch for SVGP with learned inducing locations, minibatching, and stable linear algebra [5]. The code enforces the protocol outlined above: replication-specific standardization, a fixed test grid, latent- f prediction only, and the automatic construction of the $R \times J$ coverage matrix.

4. Coverage Results

4.1. Pointwise Coverage and the Monotone α Trend

Figure 2 shows the pointwise coverage curve for baseline model ($\alpha = 1$). For approximately half of the points in testing range $[-2.5, 2.5]$, pointwise coverages are visibly lower than nominal target 0.95 line. This validates our initial hypothesis that standard SVGP produces undercoverage very often.

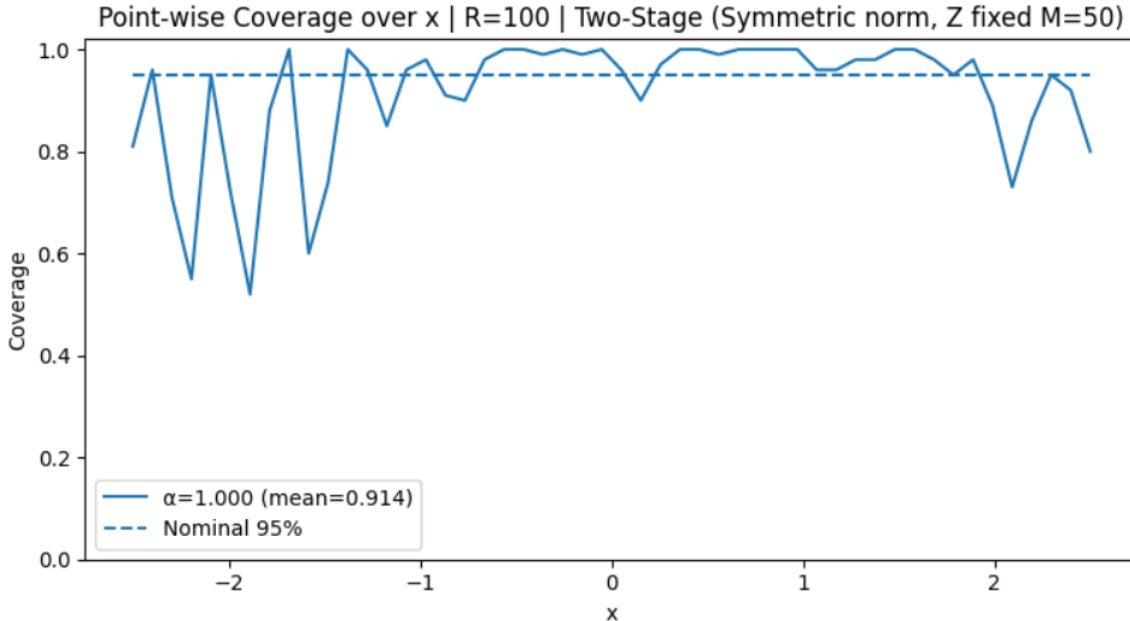


Figure 2: Baseline pointwise coverage ($\alpha = 1$)

Figure 3 shows the pointwise coverage curves for different tempering levels $\alpha \in \{1, 0.8, 0.6, 0.5, 0.3, 0.1, 0.0\}$. As α decreases, the coverage range monotonically increases and quickly approaches 1 throughout the entire test range. This monotonic improvement is exactly what our *two-stage* design predicted: the first stage (standard SVGP) learns the mean function and kernel hyperparameters, and the second stage (TVB with $\beta=1/\alpha$) only increases the variational covariance of

$q(u)$. Since the second stage keeps the mean and kernel fixed, any widening of the coverage must come from increased posterior uncertainty, not from extra randomness or error due to re-fitting. This is exactly what we observe in our experiments.

We also observe that most undercoverage occurs near the test-range boundaries; see Appendix [subsection A.1](#) for an explanation.

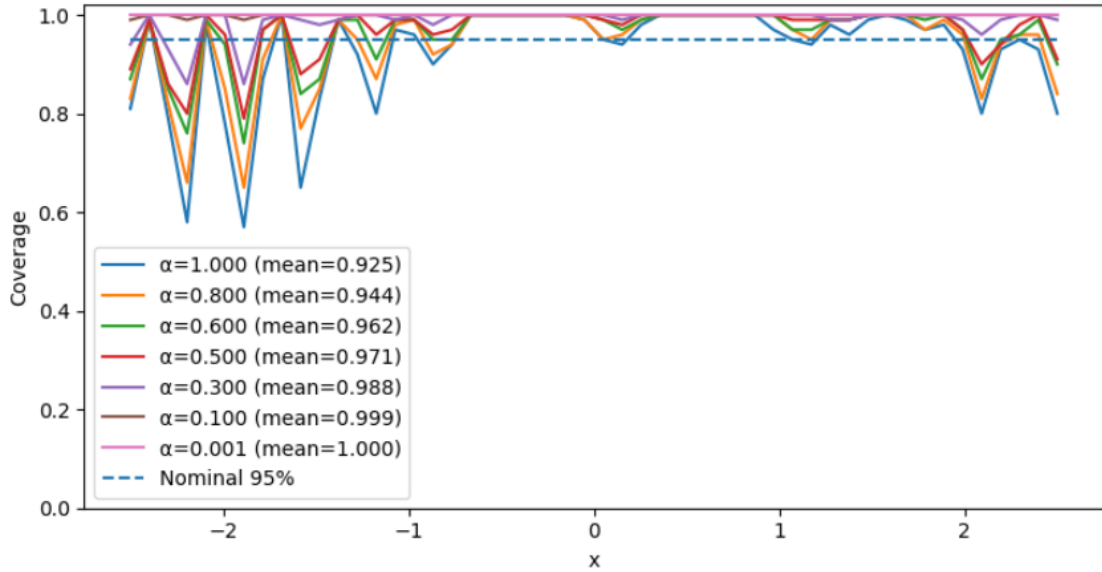


Figure 3: Pointwise coverage over x (latent f) across α . Curves rise monotonically as $\alpha \downarrow$, validating that Stage-2 tempering acts primarily by widening credible bands. Dashed line: nominal 0.95.

4.2. Stage-1 is accurate; Stage-2 widens bands.

Several diagnostics confirm that Stage-1 already fits f_0 well and that Stage-2 primarily changes uncertainty, not the mean: (i) the “fit vs truth” bias curves barely move across α as seen in [Figure 4](#), while (ii) the band $1.96 \text{std}(x)$ lifts monotonically with the largest increase at the boundaries, where $\text{std}(x)$ is the model’s posterior uncertainty (one standard deviation) for the latent function value at location x (see [Figure 11](#) and [Figure 5](#)). (iii) Moreover, the negative ELBO loss converges to a local minimum, which indicates an accurate prediction of the latent function f ([Figure 6](#)). Together these imply we are *calibrating* without degrading the mean fit.

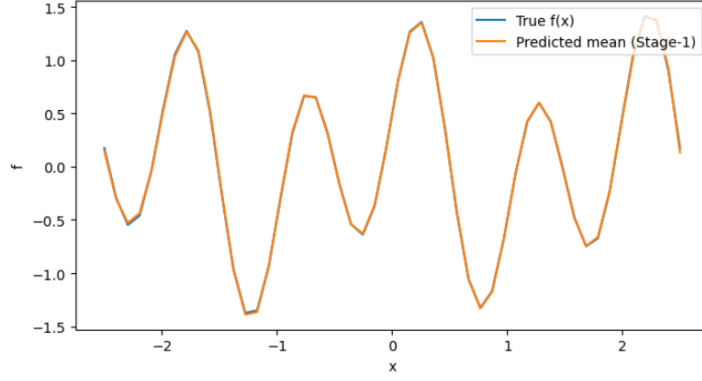


Figure 4: The Predicted fit curve nearly overlaps with the true function.

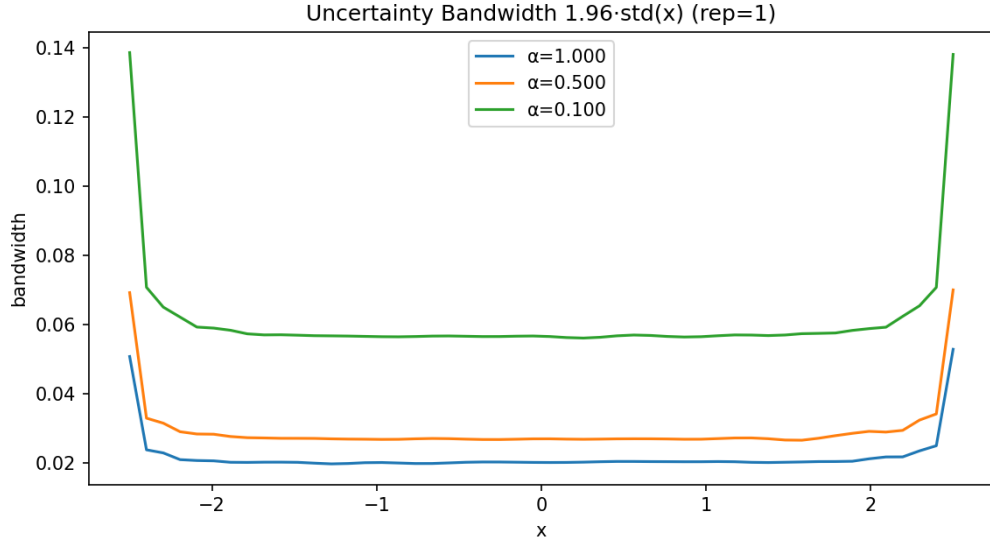


Figure 5: Overlay of $1.96 \cdot \text{std}(x)$ for $\alpha \in \{1.0, 0.5, 0.1\}$. The envelope thickens monotonically as $\alpha \downarrow$, with the largest lift occurring near boundaries.

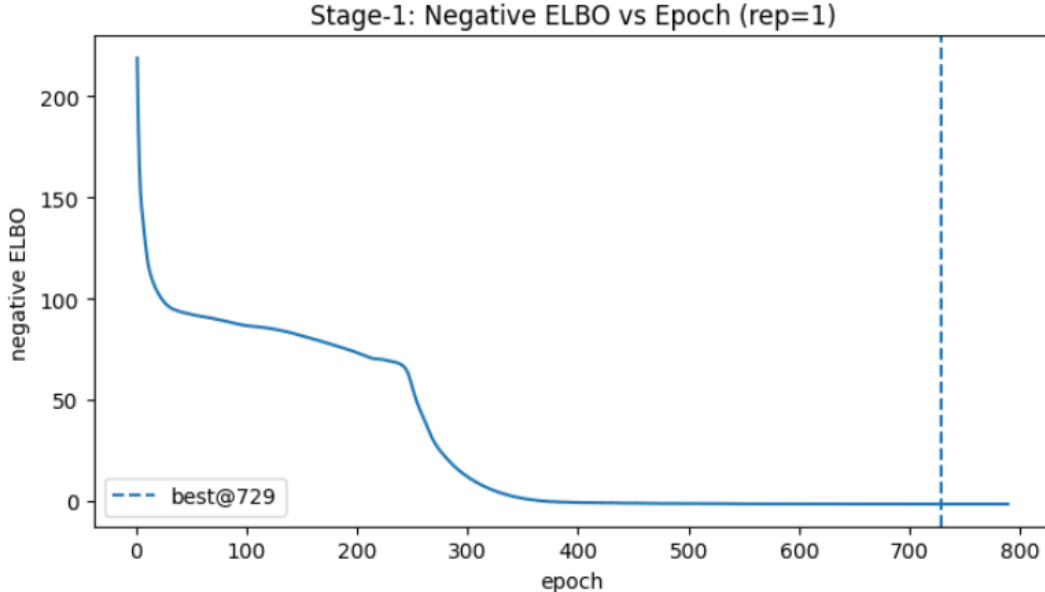


Figure 6: The negative ELBO loss curve converges to a local minimum typically around 1200 epochs, landing around 1.5.

4.3. Evaluating α values

Table 2 summarizes the performance of the main diagnostics across different α values. Mean latent- f coverage increases monotonically as α decreases, while the mean MSE remains essentially constant, indicating that tempering expands the uncertainty bands without degrading point estimation accuracy. The mean negative ELBO and its standard deviation evolve smoothly with $\beta = 1/\alpha$, showing no signs of numerical instability or optimization failure. Taken together, these checks suggest that the increase in coverage is driven by the intended covariance calibration in Stage 2, rather than by parameter drift or training instability.

Table 2: Summary of calibration diagnostics over replications for each α in the grid.

α	mean cov.	MSE_{mean}	$-\text{ELBO}_{\text{mean}}$	$-\text{ELBO}_{\text{sd}}$
1.0	0.9252	2.09×10^{-4}	-1.70	3.01×10^{-2}
0.8	0.9440	2.09×10^{-4}	-1.64	3.15×10^{-2}
0.6	0.9618	2.09×10^{-4}	-1.55	3.39×10^{-2}
0.5	0.9708	2.09×10^{-4}	-1.47	3.58×10^{-2}
0.3	0.9882	2.09×10^{-4}	-1.18	4.38×10^{-2}
0.1	0.9994	2.09×10^{-4}	0.21	8.56×10^{-2}
0.001	1.0000	2.09×10^{-4}	1.54×10^2	6.69

5. Calibration methodology

The goal now is to find the best *pointwise* tempering level α_j^* at each test point x_j^* , and thereby construct an $\boldsymbol{\alpha}^*$ vector that improves pointwise coverage toward the nominal level $(1 - \gamma)$ across the entire test grid. Here, $\gamma \in (0, 1)$ is the nominal *miscoverage* level, so $(1 - \gamma)$ is the target (nominal) pointwise coverage; for example, $\gamma = 0.05$ corresponds to 95% credible bands.

This section is organized as follows. In [subsection 5.1](#), we introduce the two-split bootstrap design and define the pseudo-truth on the fixed test grid $\mathbf{X}^*$. In [subsection 5.2](#), we describe how Split 1 constructs $\tilde{\mathbf{f}}^{\text{truth},(r)}$ using a baseline SVGP with $\alpha = 1$. In [subsection 5.3](#), we detail the Split 2 bootstrap-based calibration procedure, including the Stage-1 base fit and the Stage-2 α -tempered covariance refits that produce pointwise empirical coverage curves $\alpha \mapsto \bar{c}_j(\alpha)$ that is similar to the previous coverage assessment. Finally, in [subsection 5.4](#), we define the resulting $\boldsymbol{\alpha}^*$ -dictionary, which records the calibrated tempering level at each test location for each replication.

A high-level summary of the full calibration pipeline is given in [Algorithm 1](#). The notation matches the construction in [Sections 5.1–5.3](#).

5.1. Two-split Design and Pseudo-truth

Our calibration procedure is constructed based on a repetitive two-split design, which employs a controllable randomization method. The goal is to quantify and then correct the undercoverage of the α -tempered SVGP’s nominal $(1 - \gamma)$ credible bands for the latent function f at a fixed test grid $\mathbf{X}^* = \{x_j^*\}_{j=1}^{n_{\text{test}}}$, we fix the test grid size at $n_{\text{test}} = 50$.

For each Monte Carlo replication $r = 0, 1, \dots, R - 1$ we proceed as follows: We start from a fixed *base seed* s_0 and define the replication seed

$$s_r = s_0 + r.$$

This seed is used to initialize all the random sources in the replication process: generating the stratified one-dimensional design of `[train_low, train_high]`, the random permutation that randomly divides the training data into two halves, and the index of each bootstrap sample extracted from the second half part with replacement. In the implementation, the `torch.Generator` with the seed s_r generates all these random quantities. This controlled randomization ensures that the replication is independent, reproducible, and comparable.

Given the seed value s_r , we use the same data generation mechanism as in the baseline

coverage study to generate the training set.

$$\mathcal{D}^{(r)} = \{(x_i^{(r)}, y_i^{(r)})\}_{i=1}^{n_{\text{train}}}, \quad y_i^{(r)} = f_0(x_i^{(r)}),$$

The test grid $\mathbf{X}^*$ and the inducing point positions Z (after symmetrical linear scaling in the input domain) are *deterministic* and remain unchanged in all repeated experiments; they are not dependent on the seed. Therefore, any differences in the coverage range in different repeated experiments are entirely due to the random training design and bootstrap sampling.

We then draw a random permutation of the indices $\{1, \dots, n_{\text{train}}\}$ using the replication seed s_r and form an equal two-way split

$$\mathcal{D}_1^{(r)} \cup \mathcal{D}_2^{(r)} = \mathcal{D}^{(r)}, \quad |\mathcal{D}_1^{(r)}| = |\mathcal{D}_2^{(r)}| = n_{\text{train}}/2,$$

with disjoint index sets determined by this permutation. According to section 2, we fix $n_{\text{train}} = 500$.

The *first split* $\mathcal{D}_1^{(r)}$ is used once, to fit a baseline SVGP with $\alpha = 1$ and to construct an approximation to the latent truth with lowest accuracy loss. Evaluating the fitted latent process at $\mathbf{X}^*$ yields a vector

$$\tilde{\mathbf{f}}^{\text{truth},(r)} = (\tilde{f}_1^{\text{truth},(r)}, \dots, \tilde{f}_{n_{\text{test}}}^{\text{truth},(r)}),$$

which we treat as a *pseudo-truth* for replication r . The *second split* $\mathcal{D}_2^{(r)}$ is reserved entirely for calibration: we first fit a Stage-1 base SVGP with $\alpha = 1$ on $\mathcal{D}_2^{(r)}$ and then, for each α in a grid $\mathcal{A}$, repeatedly refit the variational covariance on bootstrap resamples of $\mathcal{D}_2^{(r)}$ to estimate how often the α -tempered credible bands contain $\tilde{\mathbf{f}}^{\text{truth},(r)}$.

At a high level, the calibration pipeline for each replication therefore consists of four steps:

1. **Randomised data generation and split** under seed s_r ;
2. **Split-1 baseline fit** to build a pseudo-truth at $\mathbf{X}^*$;
3. **Split-2 baseline fit** to obtain a stable Stage-1 SVGP on $\mathcal{D}_2^{(r)}$;
4. **Bootstrap-based α -tempered covariance refits** on $\mathcal{D}_2^{(r)}$ to estimate pointwise coverage curves $\alpha \mapsto \bar{c}_j(\alpha)$ at each test location.

The following sections provide detailed explanations of the functions of Split 1 and Split 2, the complete algorithm flow, and how these coverage curves are aggregated into a ” α dictionary” at a specific position, which records the calibration tempering level of each test point.

5.2. Split 1: Pseudo-truth Construction

On the first split $\mathcal{D}_1^{(r)}$ we fit a baseline SVGP with $\alpha = 1$, using the same model and training protocol as in the earlier coverage study:

- Constant mean function and RBF covariance with output scale;
- Fixed inducing points Z on the standardised input domain;
- Gaussian likelihood with a small nugget;
- Mini-batch Adam optimisation with early stopping based on the negative ELBO.

Let $\tilde{f}_1^{(r)}$ denote the resulting latent process. We evaluate its posterior mean at the test grid

$$\tilde{f}_j^{\text{truth},(r)} = \mathbb{E}[\tilde{f}_1^{(r)}(x_j^*) \mid \mathcal{D}_1^{(r)}], \quad j = 1, \dots, J,$$

and treat the vector

$$\tilde{\mathbf{f}}^{\text{truth},(r)} = (\tilde{f}_1^{\text{truth},(r)}, \dots, \tilde{f}_J^{\text{truth},(r)})$$

as a *pseudo-truth* for replication r .

Using $\tilde{\mathbf{f}}^{\text{truth},(r)}$ instead of the analytical $f_0(\mathbf{X}^*)$ deliberately mimics the realistic Kriging setting where the true latent field is unknown: coverage is assessed relative to what a well-fitted “gold-standard” GP would report, rather than relative to the simulator itself.

By using the $\tilde{\mathbf{f}}^{\text{truth},(r)}$ instead of the analytical function $f_0(\mathbf{X}^*)$, we are simulating the actual Kriging setup, where the true underlying field is unknown: the coverage assessment is based on the values reported by a well-fitted “gold-standard” generalized normal distribution, rather than relative to the simulator itself.

5.3. Split 2: Bootstrap-based Calibration

The second split $\mathcal{D}_2^{(r)}$ is entirely dedicated to calibration. It follows the Two-Stage process of coverage assessment: the “basic” fitting in the first stage and multiple second-stage α -tuning and covariance re-fitting based on bootstrap sampling.

The second split $\mathcal{D}_2^{(r)}$ is devoted entirely to calibration. It follows the Two-Stage procedure in coverage assessment: a Stage-1 “base” fit and many Stage-2 α -tempered covariance refits on bootstrap resamples.

Stage–1 Base Model on $\mathcal{D}_2^{(r)}$

We first fit a standard SVGP ($\alpha = 1$) on $\mathcal{D}_2^{(r)}$ using the same architecture, inducing points Z , and training settings as above. This produces a parameter state

$$(\hat{\theta}_{S_2}^{(r)}, \hat{\phi}_{S_2}^{(r)}),$$

where

$$\hat{\theta}_{S_2}^{(r)} = (\hat{\ell}_{S_2}^{(r)}, \widehat{\sigma_{fS_2}^2}^{(r)}), \quad \hat{\phi}_{S_2}^{(r)} = (\widehat{\sigma^2}_{S_2}^{(r)}, \hat{Z}_{S_2}^{(r)}, \hat{\mu}_{S_2}^{(r)}, \hat{S}_{S_2}^{(r)}).$$

Here, $\hat{\theta}_{S_2}^{(r)}$ contains the kernel hyperparameters, and $\hat{\phi}_{S_2}^{(r)}$ collects the likelihood noise variance, inducing locations, and variational parameters in $q(u) = \mathcal{N}(\mu, S)$. The Stage–1 base model is used only for initialization and for learning a stable variational mean and kernel structure at $\alpha = 1$ on $\mathcal{D}_2^{(r)}$.

Stage–2 α -tempered Covariance Refits.

Let $\mathcal{A} = \{\alpha_1, \dots, \alpha_A\} \subset (0, 1]$ be a prespecified grid of tempering parameters. We fix $A = 100$ and choose equally spaced values in the range $[0.1, 1]$, i.e.

$$\alpha_a = 0.1 + \frac{a-1}{A-1}(1-0.1), \quad a = 1, \dots, A.$$

For each $\alpha \in \mathcal{A}$ and each bootstrap index $b = 1, \dots, B$, we perform the following steps:

1. **Bootstrap sample.** Form a bootstrap resample $\mathcal{D}_{2,b}^{(r)}$ by sampling $|\mathcal{D}_2^{(r)}|$ points from $\mathcal{D}_2^{(r)}$ with replacement.
2. **Reset to Stage–1 base.** Reset the SVGP parameters to $(\hat{\theta}_{S_2}^{(r)}, \hat{\phi}_{S_2}^{(r)})$.
3. **Freeze all but the variational covariance.** We freeze all kernel hyperparameters, the inducing–point locations, the variational mean of $q(u)$, and the likelihood noise. Only covariance–related parameters of the inducing–point variational distribution (e.g. the Cholesky factor of its covariance) are left trainable.
4. **Optimise the α -tempered ELBO.** On $\mathcal{D}_{2,b}^{(r)}$ we maximise the α -tempered variational objective

$$\mathcal{L}_\alpha(q) = \alpha \mathbb{E}_{q(f)}[\log p(y_{2,b}^{(r)} | f)] - \text{KL}(q(u) \| p(u)),$$

where $\log p(y_{2,b}^{(r)} | f)$ denotes the log-likelihood of all observations in the bootstrap sample $\mathcal{D}_{2,b}^{(r)}$. Smaller α yield heavier-tailed posteriors and wider credible bands. Optimization uses Adam on the covariance parameters only.

5. **Construct pointwise credible bands.** After convergence, we evaluate the latent posterior at the test grid, obtaining a mean–variance pair $(\mu_j^{(\alpha,b,r)}, (\sigma_j^{(\alpha,b,r)})^2)$ for each x_j^* . The nominal $(1 - \gamma)$ Gaussian credible interval for the latent function is

$$C_j^{(\alpha,b,r)} = \left[\mu_j^{(\alpha,b,r)} - z_{1-\gamma/2} \sigma_j^{(\alpha,b,r)}, \mu_j^{(\alpha,b,r)} + z_{1-\gamma/2} \sigma_j^{(\alpha,b,r)} \right],$$

with $z_{1-\gamma/2}$ the standard Normal quantile (e.g. 1.96 for 95% bands).

6. **Record pointwise inclusion indicators.** Using the pseudo-truth from Split 1, we record for each test location

$$H_j^{(\alpha,b,r)} = \mathbb{I}\{\tilde{f}_j^{\text{truth},(r)} \in C_j^{(\alpha,b,r)}\}, \quad j = 1, \dots, J.$$

For a fixed replication r and α , stacking these indicators over $b = 1, \dots, B$ yields a matrix

$$H^{(\alpha,r)} \in \{0, 1\}^{B \times J},$$

and the *bootstrap estimate of pointwise coverage* at x_j^* is

$$\hat{c}_j^{(\alpha,r)} = \frac{1}{B} \sum_{b=1}^B H_j^{(\alpha,b,r)}.$$

We repeat the entire process for all $\alpha \in \mathcal{A}$ and for all replications $r = 1, \dots, R$. By averaging the results of the repeated trials, we can obtain the approximate value of the Monte Carlo method’s coverage range curve at each test point:

$$\bar{c}_j(\alpha) = \frac{1}{R} \sum_{r=1}^R \hat{c}_j^{(\alpha,r)}, \quad j = 1, \dots, J, \quad \alpha \in \mathcal{A}.$$

5.4. α -dictionary: Summarize Calibrated Tempering Levels

The output of Algorithm 1 is, for each test location x_j^* , an empirical coverage curve $\alpha \mapsto \bar{c}_j(\alpha)$ defined on the grid $\mathcal{A}$. In our experiments these curves are approximately monotone in α : for $\alpha = 1$ we typically observe undercoverage, while sufficiently small α values produce overcoverage.

Given a target coverage level $1 - \gamma$ (e.g. 0.95), we define a *pointwise calibrated tempering level* at each test location by

$$\alpha_j^* = \arg \min_{\alpha \in \mathcal{A}} |\bar{c}_j(\alpha) - (1 - \gamma)|, \quad j = 1, \dots, J.$$

This collection

$$\boldsymbol{\alpha}^* = (\alpha_1^*, \dots, \alpha_J^*)$$

forms an α -*dictionary* on the test grid: at position x_j^* , it tells us how much the posterior probability distribution of SVGP needs to be adjusted (relative to the standard $\alpha = 1$ model) to achieve the specified accuracy in the coverage range. In the Calibration Results section 6, we will present the coverage curve $\bar{c}_j(\alpha)$ and the resulting dictionary $\boldsymbol{\alpha}^*$, and study how the value of the dictionary α_j^* changes with spatial position and the local difficulty of the prediction task.

6. Calibration Results

With the α dictionary, we select a pointwise calibrated weight $\alpha^*(x_j)$ by matching the target nominal 95% coverage as closely as possible,

$$\alpha_j^* = \arg \min_{\alpha \in \mathcal{A}} |\bar{c}_j(\alpha) - (1 - \gamma)|, \quad j = 1, \dots, J.$$

For test points that are already over-covered at $\alpha = 1$ (i.e. $c_j(1) \geq 0.95$ for all $\alpha \leq 1$), we explicitly set $\alpha^*(x_j) = 1$ to avoid further inflating uncertainty. This yields a vector $\boldsymbol{\alpha}^* = (\alpha^*(x_1), \dots, \alpha^*(x_J))$ that varies smoothly over most of the domain but takes markedly smaller values in regions where the baseline SVGP exhibits severe undercoverage. Table 3 shows selected α^* for each test point x_j .

Table 3: Pointwise calibrated tempering levels $\alpha^*(x_j^*)$ on the fixed test grid $\mathcal{X}^*$.

x_j^*	α^*	x_j^*	α^*	x_j^*	α^*	x_j^*	α^*	x_j^*	α^*
-2.500	0.15	-1.480	0.42	-0.459	1.00	0.561	1.00	1.582	1.00
-2.398	1.00	-1.378	0.67	-0.357	1.00	0.663	1.00	1.684	1.00
-2.296	0.43	-1.276	1.00	-0.255	1.00	0.765	1.00	1.786	1.00
-2.194	0.26	-1.173	1.00	-0.153	1.00	0.867	1.00	1.888	1.00
-2.092	0.70	-1.071	0.70	-0.051	1.00	0.969	1.00	1.990	0.76
-1.990	1.00	-0.969	0.97	0.051	1.00	1.071	1.00	2.092	0.51
-1.888	0.41	-0.867	1.00	0.153	1.00	1.173	1.00	2.194	1.00
-1.786	0.54	-0.765	0.87	0.255	1.00	1.276	0.96	2.296	1.00
-1.684	1.00	-0.663	0.99	0.357	1.00	1.378	1.00	2.398	1.00
-1.582	1.00	-0.561	1.00	0.459	1.00	1.480	1.00	2.500	0.34

In order to evaluate the performance of the calibrated schedule $\boldsymbol{\alpha}^*(x)$ in a fresh experiment, we re-input it into the same two-stage SVGP process used in coverage assessment before.

Let $\alpha^*(x_j^*)$ denote the calibrated tempering level at test location x_j^* , and let

$$\mathcal{A}^* = \text{unique}\{\alpha^*(x_1^*), \dots, \alpha^*(x_{50}^*)\}$$

be the set of distinct calibrated values on the grid. For each distinct value $\alpha^* \in \mathcal{A}^*$, we conduct a separate double-check experiment: we rerun the two-stage process with the temperature control parameter $\beta = 1/\alpha^*$, and at the end, we only record the 95% prediction coverage at the test positions x_j^* where $\alpha^*(x_j^*) = \alpha^*$. This operation is performed for all $\alpha^* \in \mathcal{A}^*$, and the results are averaged after repeating the experiment for $R = 100$ replications. Eventually, an empirical double-check coverage vector of dimension 50 is obtained, corresponding to the pointwise schedule $\alpha^*(x)$.

Figure 7 shows the plotted empirical "double-check" coverages over x_j .

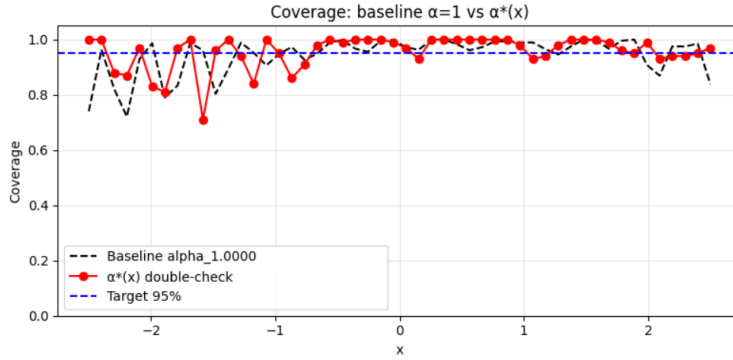


Figure 7: Calibrated double-check coverage & Baseline coverage

The recheck results indicate a clear improvement compared to the baseline $\alpha = 1$ coverage range. In many problematic test locations (particularly on the left side of the test area), the baseline coverage was within the range of 0.7–0.9, while the calibrated $\alpha^*(x)$ brought these values closer to the target level of 0.95. In the central and right areas, since the baseline coverage was already close to the standard value, the calibrated curve remained largely unchanged. A small portion of the test points were still not covered, however, had an empirical coverage ranging around 0.7–0.85.

To understand these remaining discrepancies, we compared the coverage of the double-check with the uncertainty in the original calibration estimates. Figure 8 plots the calibration mean and its ± 1 standard deviation band (grey/black) at each test location under $\alpha^*(x)$ and the double-check coverage (red dots) under the same $\alpha^*(x)$. Most of the red dots fall well within the grey band and are close to the nominal 0.95 line, which confirms that the calibrated $\alpha^*(x)$ at problematic locations makes the coverage closer to the target, while leaving already well-behaved regions essentially unchanged.

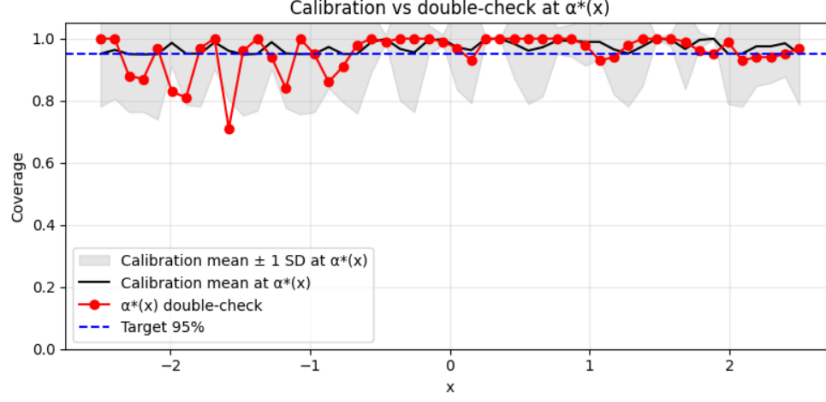


Figure 8: Calibrated double-check coverage & Baseline coverage

Figure 9 further links the remaining differences with the variability of the calibration estimates. For each test position, we extract from the α -dictionary the average calibration coverage at $\alpha^*(x)$ and its standard deviation across different calibration replications, and plot the double-check deviation that differs from 0.95 along with this standard deviation. In the double-check experiments, almost all the "bad" points fall within the calibration range given by mean $\pm$ one standard deviation, while those locations that still show clear insufficient coverage are precisely those with the largest calibration standard deviations (typically between 0.15–0.2). This indicates that the actual coverage ranges substantially across training sets: these failure cases are consistent with extracting an unlucky repeated sample from a very noisy distribution, rather than revealing systematic defects in the calibration scheme.

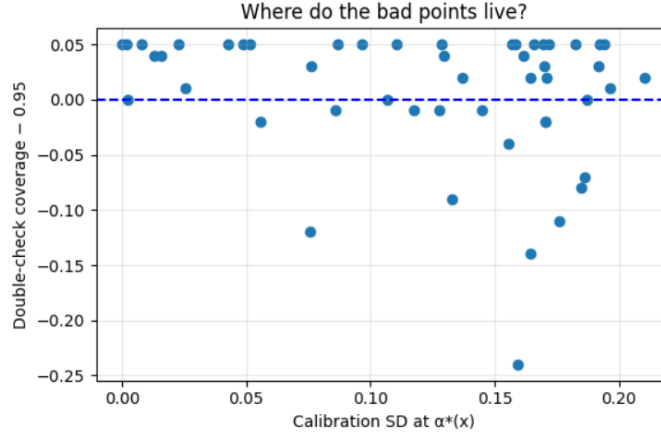


Figure 9: Calibrated double-check coverage & baseline coverage

Overall, the α -dictionary and the induced $\alpha^*(x)$ indicate that merely adjusting the variational covariance can improve the coverage of those regions where the baseline SVGP lacks

confidence, while having little effect on the regions with good performance. The double-test experiments in Figures 7, 8, and 9 support this view: most test points are closer to the nominal 95% level, and the remaining uncovered areas can be explained by the intrinsic variability estimated by calibration, rather than being caused by the structural limitations of the method.

7. Discussion

This study aims to establish a practical procedure for calibrating the uncertainty in SVGPs using TVB objectives. Our goal is twofold. Firstly, to address the deficiency in the credible interval coverage of standard SVGP while maintaining its computational efficiency and inducing point structure. Secondly, to conduct a preliminary coverage study on synthetic data with known true values f_0 , in order to achieve the first milestone. The benchmark SVGP ($\alpha = 1$) has a systematic under-coverage of the latent function f on a fixed evaluation grid (Figure 3). Reducing the α value gradually expands the posterior interval and increases the actual coverage monotonically, making it closer to the nominal level (Figures 3 and 4). These research results indicate that adjusting the likelihood function appropriately can enhance calibration performance without altering the modeling class or inducing-point mechanism [2, 3, 5].

The primary message of our project is that calibration can be modified exclusively via a single scalar temperature within the SVGP loop. This is conceptually consistent with the generalized Bayesian update, where the posterior is constructed by weighting the likelihood [7]. It is also consistent with broader observations that variational methods optimized under the $\text{KL}(q||p)$ objective often generate overly concentrated posteriors [4]. In our setup, the α -tempered ELBO reduces the weight of the likelihood and counteracts overconfidence, thereby leading to higher empirical coverage of the latent function. Our conclusion from Figures 3, 4, and Table 2 is that SVGP is both fast and accurate regarding the mean but lacks sufficient coverage, while TVB consistently increases coverage to meet the target level.

Another contribution of this study lies in the introduction of clear calibration steps, which construct an α -dictionary and a pointwise calibrated tempering profile $\alpha^*(x)$. Through a two-split bootstrap method, we calculated the point-based coverage curve $\alpha \mapsto c_j(\alpha)$ for each test position x_j^* and for each α value (in a dense grid), and then selected $\alpha^*(x_j^*)$ that estimated the coverage level closest to the nominal 95% level. The resulting dictionary (Table 3) shows smooth variations in most areas and takes smaller values on the left (i.e., the regions where the baseline SVGP has the least confidence). Reinserting this $\alpha^*(x)$ into the original two-stage SVGP pipeline and repeating the entire experiment yields a "double-

check” coverage curve, which is visibly better than the baseline $\alpha = 1$ at many problematic locations and remains almost unchanged in areas where the original coverage was already close to the nominal level (Figure 8). The diagnostics in Figure 9 further indicate that the few points that still remain uncovered after calibration are precisely those with the largest calibration standard deviations, suggesting that their apparent failure is driven by high sampling variability rather than structural defects in the calibration.

From a practical perspective, these results suggest that an α -tempered SVGP with a carefully constructed calibration step can deliver scalable spatial prediction with more trustworthy uncertainty quantification. In applications such as geostatistics and spatial risk assessment, practitioners often require both accurate point predictions from nearby measurements and predictive intervals that meaningfully reflect uncertainty. Our pipeline keeps the inducing-point machinery and mini-batch training of SVGP, but augments it with a lightweight TVB layer that adjusts posterior spread. With a well-chosen $\alpha^*(x)$, the resulting model offers interpretable, location-specific tempering: in regions where the baseline SVGP is already well-calibrated, $\alpha^*(x) \approx 1$, while in challenging regions, smaller $\alpha^*(x)$ visibly improve coverage without sacrificing mean accuracy (Table 2).

There are still some limitations. Firstly, all the experiments were conducted on one-dimensional, noise-free synthetic data with an RBF kernel function. Although this controlled environment eliminates the calibration effect, the actual spatial problems usually involve higher-dimensional inputs, measurement noise, non-stationary structures, and inappropriate covariance families. Extending the TVB-SVGP calibration process to noisy observations y , multi-dimensional inputs, and more diverse kernel classes (e.g., Matérn or non-stationary kernels) is an important step for the next stage. Secondly, the current α -dictionary is constructed based on a fixed evaluation grid and relies on the self-generated pseudo-truth from split 1: adapting $\alpha^*(x)$ to a new location or a completely new dataset, either an interpolation rule is needed, or an additional calibration run is required. Thirdly, although our diagnostics indicates that softening mainly changes the covariance rather than the mean, we have not systematically quantified the trade-off relationship between improving coverage and potential loss in predictive efficiency in more complex situations [10]. Finally, our work is empirical: although it is inspired by the generalized Bayesian theory and the latest theories on tempered and θ -variational methods [8, 9], whether the α^* selected through the two-split bootstrap provides formal guarantees remains an open theoretical issue.

Overall, our results validate the presence of under-coverage in standard SVGP and support a simple, theory-grounded modification via α -tempered variational Bayes. By combining a scalable two-stage SVGP with an α -dictionary and double-check calibration, we constructed

a pipeline that produces accurate predictions with empirically controllable uncertainty at a large scale. We view this as a promising foundation for future work on TVB-calibrated Kriging in higher dimensions, on real spatial data sets, and under more challenging forms of model misspecification.

Algorithm 1: Two-split SVGP Calibration Pipeline

Input: Number of replications R ; bootstrap size B ; alpha grid $\mathcal{A} = \{\alpha_1, \dots, \alpha_A\}$; test

grid $\mathcal{X}^* = \{x_j^*\}_{j=1}^J$.

Output: Coverage curves $\bar{c}_j(\alpha) = \frac{1}{R} \sum_{r=1}^R \hat{c}_j^{(\alpha,r)}$ for $j = 1, \dots, J$ and all $\alpha \in \mathcal{A}$.

for $r = 0, 1, \dots, R - 1$ **do**

 Set replication seed $s_r = s_0 + r$.

 Initialise PRNGs.

 Generate training data $\mathcal{D}^{(r)}$ from the data-generating mechanism.

 Using seed s_r , draw a random permutation of indices.

 Split $\mathcal{D}^{(r)}$ into equal halves $\mathcal{D}_1^{(r)}$ and $\mathcal{D}_2^{(r)}$.

 // Split 1: pseudo-truth

 Fit baseline SVGP ($\alpha = 1$) on $\mathcal{D}_1^{(r)}$.

 Compute pseudo-truth at the test grid $\tilde{f}_j^{\text{truth},(r)} = \mathbb{E}[\tilde{f}_1^{(r)}(x_j^*) \mid \mathcal{D}_1^{(r)}]$ for $j = 1, \dots, J$.

 // Split 2: Stage-1 base model

 Fit baseline SVGP ($\alpha = 1$) on $\mathcal{D}_2^{(r)}$.

 Save Stage-1 parameter state $(\hat{\theta}_{S2}^{(r)}, \hat{\phi}_{S2}^{(r)})$.

 // Split 2: bootstrap calibration over $\mathcal{A}$

foreach $\alpha \in \mathcal{A}$ **do**

for $b = 1, \dots, B$ **do**

 Draw bootstrap sample $\mathcal{D}_{2,b}^{(r)}$ from $\mathcal{D}_2^{(r)}$ with replacement.

 Reset SVGP to $(\hat{\theta}_{S2}^{(r)}, \hat{\phi}_{S2}^{(r)})$.

 Freeze kernel, inducing points, variational mean, and noise.

 Optimise α -tempered ELBO on $\mathcal{D}_{2,b}^{(r)}$ w.r.t. variational covariance parameters only.

 Evaluate posterior at $\mathcal{X}^*$ and form $(1 - \gamma)$ credible bands $C_j^{(\alpha,b,r)}$ for all j .

 Record indicators $H_j^{(\alpha,b,r)} = \mathbb{I}\{\tilde{f}_j^{\text{truth},(r)} \in C_j^{(\alpha,b,r)}\}$ for all j .

 Compute bootstrap coverage estimates $\hat{c}_j^{(\alpha,r)} = \frac{1}{B} \sum_{b=1}^B H_j^{(\alpha,b,r)}$ for $j = 1, \dots, J$.

References

- [1] Motonobu Kanagawa, Philipp Hennig, Dino Sejdinovic, and Bharath K. Sriperumbudur. Gaussian processes and kernel methods: A review on connections and equivalences. *Foundations and Trends in Machine Learning*, 11(5-6):367–570, 2018. doi: 10.1561/22000000060.
- [2] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA, 2006. ISBN 978-0-262-18253-9.
- [3] Michalis K. Titsias. Variational learning of inducing variables in sparse gaussian processes. In *Proceedings of the 12th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 5, pages 567–574. PMLR, 2009. URL <https://proceedings.mlr.press/v5/titsias09a.html>.
- [4] David M. Blei, Alp Kucukelbir, and Jon D. McAuliffe. Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017. doi: 10.1080/01621459.2017.1285773.
- [5] Jacob R. Gardner, Geoff Pleiss, David Bindel, Kilian Q. Weinberger, and Andrew Gordon Wilson. Gpytorch: Blackbox matrix–matrix Gaussian process inference with gpu acceleration. In *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*, 2018. URL <https://gpytorch.ai/>.
- [6] Dennis Nieman, Botond Szabó, and Harry van Zanten. Contraction rates for sparse variational approximations in Gaussian process regression. *Journal of Machine Learning Research*, 23(205):1–26, 2022. URL <https://www.jmlr.org/papers/v23/21-1128.html>. PDF: <https://www.jmlr.org/papers/volume23/21-1128/21-1128.pdf>.
- [7] Pier Giovanni Bissiri, Chris Holmes, and Stephen G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):1103–1130, 2016. doi: 10.1111/rssb.12158.
- [8] Nicholas Syring and Ryan Martin. Calibrating general posterior credible regions. *Biometrika*, 106(2):479–486, 2019. doi: 10.1093/biomet/asy054.
- [9] Yun Yang, Debdeep Pati, and Anirban Bhattacharya. θ -variational inference with statistical guarantees. *The Annals of Statistics*, 48(6):3570–3598, 2020. doi: 10.1214/19-AOS1827.

- [10] Konstantinos Pitas and Julyan Arbel. The fine print on tempered posteriors. In *Proceedings of the 15th Asian Conference on Machine Learning (ACML 2023)*, pages 1–16, 2023. URL <https://arxiv.org/abs/2303.01491>. Improves calibration via tempering but may reduce predictive accuracy.

A. Appendix

A.1. Undercoverage Places in Coverage Assessment

Although the overall curve shows a uniform upward trend, the *redisual* shortfalls at $\alpha = 1$ mainly occur near the boundary. Figure 10 visualizes the $R \times J$ coverage matrix as heatmaps for $\alpha \in \{1.0, 0.5, 0.1\}$. The purple 0 pixels at $\alpha=1$ cluster at the edges and a few high-curvature spots, and they thin out at $\alpha=0.5$ before ceasing at $\alpha=0.1$. This “edge effect” is further explained by the bias-vs-bandwidth check. Figure 11 demonstrates how the absolute bias $|\mu(x) - f_0(x)|$ is relatively small within this test range, but it slightly increases near the ends. Meanwhile, the posterior bandwidth of $1.96 \text{std}(x)$ shows the most significant growth at the edges. As α decreases, the bias/bandwidth ratio will drop below 1, restoring the coverage range.

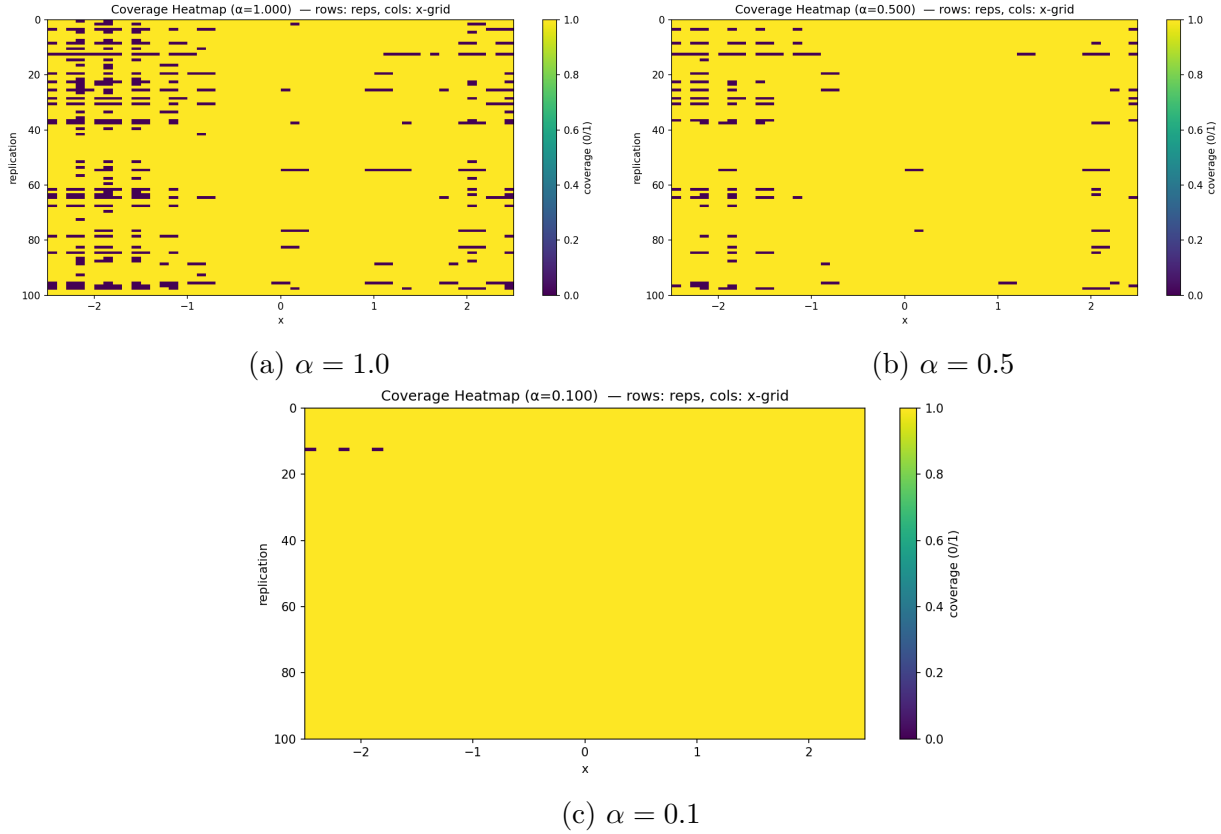


Figure 10: Coverage heatmaps ($R=100$ reps; rows=reps, cols=test x).

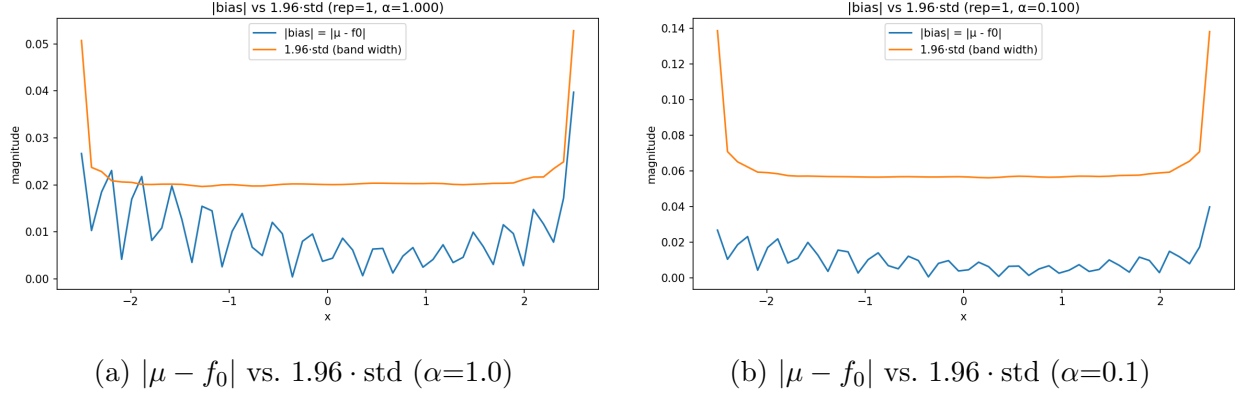


Figure 11: Bias is small and shape-stable; bands widen as $\alpha \downarrow$, especially at boundaries, resolving under-coverage without changing the mean.

A.2. Notation

Table 4: Notation used throughout the paper.

Symbol	Meaning
x	Input location (1D coordinate).
y	Observed response / training target; in this project $y_i = f_0(x_i)$ (noise-free).
f	Latent (noise-free) regression function modeled by a GP.
f_0	True data-generating function used only for synthetic evaluation.
$GP(0, k_\theta)$	Gaussian process prior with zero mean and kernel k_θ .
$k(x, x')$	Kernel/covariance function encoding spatial similarity.
$\theta = (\ell, \sigma_f^2)$	Kernel hyperparameters: length-scale ℓ and signal variance σ_f^2 .
ℓ	RBF kernel length-scale.
σ_f^2	RBF kernel signal variance (output scale).
n_{train}	Training set size (here 500).
n_{test}	Test grid size (here 50).
R	Number of Monte Carlo replications (here 100).
J	Number of test locations; used interchangeably with n_{test} (so $J = n_{\text{test}}$).
X_{train}^*	Raw (unstandardized) training inputs generated by LHS.
$X_{\text{test}}^* / \mathbf{X}^*$	Fixed raw test grid over $[-2.5, 2.5]$.

Symbol	Meaning
$X_{\text{train}}, X_{\text{test}}$	Standardized inputs after symmetric linear normalization to $[-1, 1]$.
L, H	Raw domain endpoints (here $L = -2.5$, $H = 2.5$).
$m = \frac{L+H}{2}$	Midpoint used in symmetric linear normalization. (Note: this conflicts with m used for inducing-point count; consider renaming this midpoint to c.)
$h = \frac{H-L}{2}$	Half-range used in symmetric linear normalization.
x_{std}	Standardized input: $(x_{\text{raw}} - m)/h$.
M / m	Number of inducing points (you use both notations; you set it to 50).
Z^*	Raw inducing inputs (equally spaced on $[-2.5, 2.5]$).
Z	Standardized inducing inputs (after the same normalization).
$u = f(Z)$	Inducing variables (latent function evaluated at inducing inputs).
$q(u) = \mathcal{N}(\mu, S)$	Variational posterior over inducing variables: mean vector μ and covariance matrix S .
μ	Variational mean of $q(u)$.
S	Variational covariance of $q(u)$.
$q(f^*) = \mathcal{N}(\mu^*, \Sigma^*)$	SVGP predictive posterior for latent values on the test grid.
μ^*	Predictive posterior mean of f^* on $\mathbf{X}^*$.
Σ^*	Predictive posterior covariance of f^* on $\mathbf{X}^*$.
$\text{std}(x)$	Posterior predictive standard deviation of $f(x)$ (one-sigma uncertainty), i.e., $\sqrt{\Sigma^*(x, x)}$.
K_{ZZ}	Kernel matrix $k(Z, Z)$.
K_{X^*Z}	Cross-kernel matrix $k(X^*, Z)$.
K_{ZX^*}	Cross-kernel matrix $k(Z, X^*)$.
$K_{X^*X^*}$	Kernel matrix $k(X^*, X^*)$.
ELBO	Standard evidence lower bound objective for variational inference.
$\mathcal{L}_\alpha$	α -tempered ELBO (expected log-likelihood scaled by α).
α	Tempering parameter in generalized Bayes; $\alpha \in (0, 1]$.
β	Temperature control parameter used in code, $\beta = 1/\alpha$.
γ	Nominal miscoverage level, so $(1-\gamma)$ is target coverage (e.g., $\gamma = 0.05$ for 95% bands).

Symbol	Meaning
$z_{1-\gamma/2}$	Standard Normal quantile used for $(1 - \gamma)$ Gaussian credible bands (e.g., 1.96 for 95%).
$\text{CI}_{0.95}(x_j^*)$	Pointwise 95% credible interval for the latent function at x_j^* .
$H \in \{0, 1\}^{R \times J}$	Coverage indicator matrix across replications and test locations.
H_{rj}	Indicator that $\text{CI}_{0.95}^{(r)}(x_j^*)$ covers the truth $f_0(x_j^*)$ in replication r .
$\widehat{C}(j)$	Pointwise frequentist coverage estimate (column mean of H at test index j).
$\overline{C}$	Overall coverage (grand mean of H).
$\mathcal{D}^{(r)}$	Full training dataset in replication r .
$\mathcal{D}_1^{(r)}, \mathcal{D}_2^{(r)}$	Two-way random split of $\mathcal{D}^{(r)}$ into equal halves.
$\tilde{\mathbf{f}}^{\text{truth},(r)}$	Pseudo-truth vector on $\mathbf{X}^*$ constructed from Split 1 in replication r .
$\tilde{f}_j^{\text{truth},(r)}$	Pseudo-truth at test location x_j^* in replication r .
s_0	Base seed.
s_r	Replication seed ($s_r = s_0 + r$).
$\mathcal{A} = \{\alpha_1, \dots, \alpha_A\}$	Grid of tempering parameters for calibration.
A	Number of grid points in $\mathcal{A}$ (here 100).
B	Bootstrap sample size (number of bootstrap refits per α).
$\mathcal{D}_{2,b}^{(r)}$	Bootstrap resample from $\mathcal{D}_2^{(r)}$ (index b) in replication r .
$H^{(\alpha,r)} \in \{0, 1\}^{B \times J}$	Bootstrap inclusion indicator matrix at fixed (α, r) .
$\widehat{c}_j^{(\alpha,r)}$	Bootstrap estimate of pointwise coverage at x_j^* for fixed (α, r) .
$\bar{c}_j(\alpha)$	Averaged (over R replications) pointwise calibration coverage curve at x_j^* .
$\alpha_j^* / \alpha^*(x_j^*)$	Pointwise calibrated tempering level at test location x_j^* .
$\boldsymbol{\alpha}^*$	Vector (dictionary) of calibrated tempering levels over the test grid.
$\mathcal{A}^*$	Set of distinct calibrated values appearing in $\boldsymbol{\alpha}^*$ (used for double-check runs).